LAB 4- Introduction to Kubernetes on Google Cloud using GKE

Lab Task 1 – Basic Kubernetes operations

After creating the google account, installing the google sdk and kubectl, I created a node -1 cluster- onlineboutique

```
oadewusi@oluSec:~/aiops/Assignments$ gcloud container clusters create onlineboutique \
--project=aiops4 \
--zone=europe-west4-a \
--machine-type=e2-standard-4 \
--num-nodes=1 \
--workload-pool aiops4.svc.id.goog \
--gateway-api "standard"
Note: The Kubelet readonly port (10255) is now deprecated. Please update your workloads to use the recommended alternatives. See https://cl
oud.google.com/kubernetes-engine/docs/how-to/disable-kubelet-readonly-port for ways to check usage and for migration instructions.
Note: Your Pod address range (`--cluster-ipv4-cidr`) can accommodate at most 1008 node(s).
Creating cluster onlineboutique in europe-west4-a... Cluster is being health-checked (Kubernetes Control Plane is healthy)...done.
Created [https://container.googleapis.com/v1/projects/aiops4/zones/europe-west4-a/clusters/onlineboutique].
To inspect the contents of your cluster, go to: https://console.cloud.google.com/kubernetes/workload_/gcloud/europe-west4-a/onlineboutique?
project=aiops4
kubeconfig entry generated for onlineboutique.
NAME            LOCATION        MASTER_VERSION      MASTER_IP      MACHINE_TYPE   NODE_VERSION        NUM_NODES  STATUS
onlineboutique  europe-west4-a  1.30.3-gke.1969001  34.34.95.197   e2-standard-4  1.30.3-gke.1969001  1          RUNNING
```

Also downloaded and unzipping the bookinfo file from GitHub before applying the yaml file

```
oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ kubectl apply -f platform/kube/bookinfo.yaml
service/details created
serviceaccount/bookinfo-details created
deployment.apps/details-v1 created
service/ratings created
serviceaccount/bookinfo-ratings created
deployment.apps/ratings-v1 created
service/reviews created
serviceaccount/bookinfo-reviews created
deployment.apps/reviews-v1 created
deployment.apps/reviews-v2 created
deployment.apps/reviews-v3 created
service/productpage created
serviceaccount/bookinfo-productpage created
deployment.apps/productpage-v1 created
```

kubectl get services:

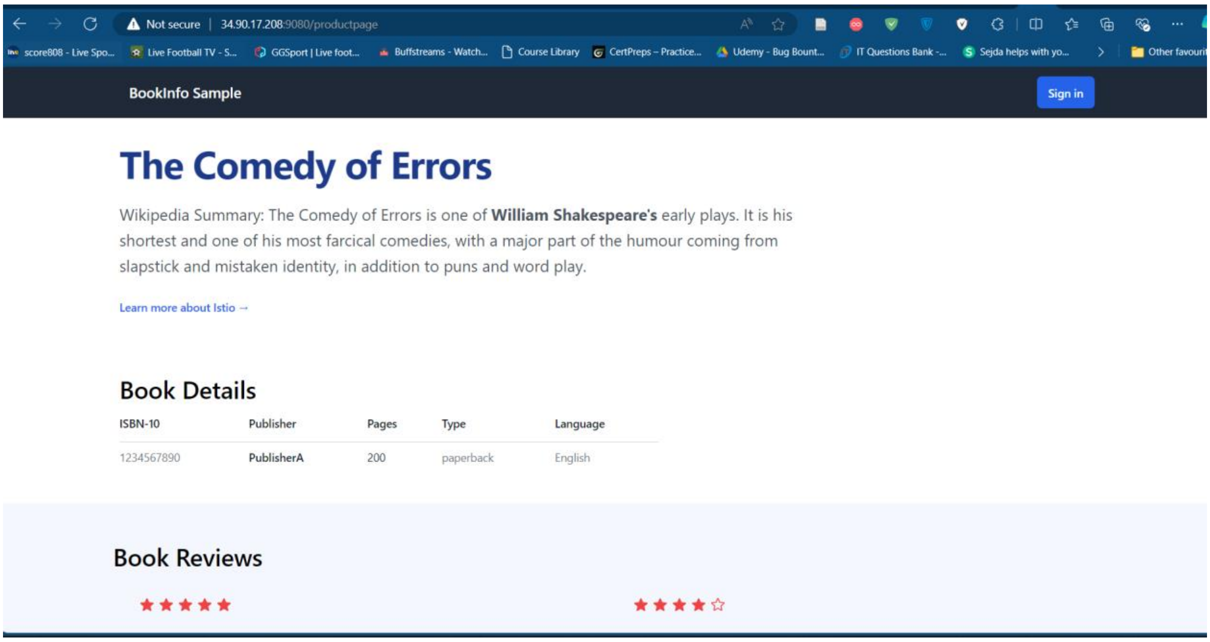
```
oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ kubectl get services
NAME          TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)    AGE
details       ClusterIP   34.118.229.26    <none>        9080/TCP   3m5s
kubernetes    ClusterIP   34.118.224.1     <none>        443/TCP    13m
productpage   ClusterIP   34.118.232.94    <none>        9080/TCP   3m
ratings       ClusterIP   34.118.235.11    <none>        9080/TCP   3m4s
reviews       ClusterIP   34.118.238.209   <none>        9080/TCP   3m2s
```

Changed the productpage to loadbalancer with : kubectl patch svc productpage -p '{"spec": {"type": "Load-Balancer"}}'

```
oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ kubectl patch svc productpage -p '{"spec": {"type": "LoadBalancer"}}'
service/productpage patched
```

```
oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ kubectl get services
NAME          TYPE           CLUSTER-IP       EXTERNAL-IP    PORT(S)          AGE
details       ClusterIP      34.118.229.26    <none>         9080/TCP         6m1s
kubernetes    ClusterIP      34.118.224.1     <none>         443/TCP          16m
productpage   LoadBalancer   34.118.232.94    34.90.17.208   9080:32237/TCP   5m56s
ratings       ClusterIP      34.118.235.11    <none>         9080/TCP         6m
reviews       ClusterIP      34.118.238.209   <none>         9080/TCP         5m58s
```

Accessed it via my web browser with its public IP:



Kubernetes Services: Services act as a stable entry point for accessing the application's backend. They provide a single, well-defined IP address and port combination that can be used to route traffic to the appropriate backend pods. Services are responsible for load balancing and forwarding incoming requests to the correct pods.

Key Functions:
Load Balancing: Distributes traffic across multiple Pods.
Service Discovery: Provides a consistent way for clients to connect to Pods.
Types of Services:
ClusterIP: Exposes the service on an internal IP in the cluster.
NodePort: Exposes the service on each Node's IP at a static port.
LoadBalancer: Provisions an external load balancer (as seen in your setup).
Commands:
List services: kubectl get services
Describe a service: kubectl describe service <service-name>

```
oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ kubectl get services
NAME          TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)    AGE
details       ClusterIP   34.118.229.26    <none>        9080/TCP   3m5s
kubernetes    ClusterIP   34.118.224.1     <none>        443/TCP    13m
productpage   ClusterIP   34.118.232.94    <none>        9080/TCP   3m
ratings       ClusterIP   34.118.235.11    <none>        9080/TCP   3m4s
reviews       ClusterIP   34.118.238.209   <none>        9080/TCP   3m2s
```

```
oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ kubectl describe svc productpage
Name:                     productpage
Namespace:                default
Labels:                   app=productpage
                          service=productpage
Annotations:              cloud.google.com/neg: {"ingress":true}
Selector:                 app=productpage
Type:                     LoadBalancer
IP Family Policy:         SingleStack
IP Families:              IPv4
IP:                       34.118.232.94
IPs:                      34.118.232.94
LoadBalancer Ingress:     34.90.17.208
Port:                     http  9080/TCP
TargetPort:               9080/TCP
NodePort:                 http  32237/TCP
Endpoints:                10.76.0.18:9080
Session Affinity:         None
External Traffic Policy:  Cluster
Events:
  Type    Reason                Age                From                   Message
  ----    ------                ----               ----                   -------
  Normal  ADD                   19m                sc-gateway-controller  default/productpage
  Normal  Type                  15m                service-controller     ClusterIP -> LoadBalancer
  Normal  UPDATE                15m (x2 over 15m)  sc-gateway-controller  default/productpage
  Normal  EnsuringLoadBalancer  15m                service-controller     Ensuring load balancer
  Normal  EnsuredLoadBalancer   15m                service-controller     Ensured load balancer
```

Deployments: Deployments are responsible for managing the lifecycle of pods, which are the actual containers running the application code. When a new version of the application needs to be deployed, the Deployment controller creates new pods with the updated code and gradually replaces the old pods. This process is known as a rolling update, ensuring that the application remains available during the update process.

Key Functions:

Scaling: Automatically adjusts the number of running Pods.

Rolling Updates: Ensures zero downtime by gradually replacing old Pods with new ones.

Self-Healing: Replaces failed Pods automatically.

Commands:

List deployments: kubectl get deployments

Describe a deployment: kubectl describe deployment <deployment-name>

```
oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ kubectl get deployments
NAME             READY   UP-TO-DATE   AVAILABLE   AGE
details-v1       1/1     1            1           26m
productpage-v1   1/1     1            1           26m
ratings-v1       1/1     1            1           26m
reviews-v1       1/1     1            1           26m
reviews-v2       1/1     1            1           26m
reviews-v3       1/1     1            1           26m
oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$
```

```
● oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ kubectl describe deployment productpage-v1
  Name:                   productpage-v1
  Namespace:              default
  CreationTimestamp:      Wed, 02 Oct 2024 12:04:55 +0200
  Labels:                 app=productpage
                          version=v1
  Annotations:            deployment.kubernetes.io/revision: 1
  Selector:               app=productpage,version=v1
  Replicas:               1 desired | 1 updated | 1 total | 1 available | 0 unavailable
  StrategyType:           RollingUpdate
  MinReadySeconds:        0
  RollingUpdateStrategy:  25% max unavailable, 25% max surge
  Pod Template:
    Labels:           app=productpage
                      version=v1
    Annotations:      prometheus.io/path: /metrics
                      prometheus.io/port: 9080
                      prometheus.io/scrape: true
    Service Account:  bookinfo-productpage
    Containers:
     productpage:
      Image:        docker.io/istio/examples-bookinfo-productpage-v1:1.20.2
      Port:         9080/TCP
      Host Port:    0/TCP
      Environment:  <none>
      Mounts:
        /tmp from tmp (rw)
    Volumes:
     tmp:
      Type:        EmptyDir (a temporary directory that shares a pod's lifetime)
      Medium:
      SizeLimit:   <unset>
    Node-Selectors:  <none>
    Tolerations:     <none>
```

Pods: Pods are the smallest deployable units in Kubernetes. They represent one or more containers that share storage and network resources. Each pod runs a single instance of the application, and multiple pods can be created to handle increased traffic or ensure high availability.

Key Functions:

Isolation: Provides an isolated environment for containers.

Networking: Each Pod gets its own IP address.

Lifecycle Management: Handles starting, stopping, and restarting containers.

Commands:

List pods: kubectl get pods

Describe a pod: kubectl describe pod <pod-name>

View logs for troubleshooting: kubectl logs <pod-name>

```
● oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ kubectl get pods
  NAME                              READY   STATUS    RESTARTS   AGE
  details-v1-6cd6d9df6b-dt9tj       1/1     Running   0          29m
  productpage-v1-57ffb6658c-x4gzd   1/1     Running   0          29m
  ratings-v1-794744f5fd-fhmqb       1/1     Running   0          29m
  reviews-v1-67896867f4-j6clg       1/1     Running   0          29m
  reviews-v2-86d5db4bd6-2rdsx       1/1     Running   0          29m
  reviews-v3-77947c4c78-mhmh9       1/1     Running   0          29m
```

```
● oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ kubectl describe pods productpage-v1-57ffb6658c-x4gzd
  Name:             productpage-v1-57ffb6658c-x4gzd
  Namespace:        default
  Priority:         0
  Service Account:  bookinfo-productpage
  Node:             gke-onlineboutique-default-pool-35c278d7-2wf5/10.164.0.3
  Start Time:       Wed, 02 Oct 2024 12:04:55 +0200
  Labels:           app=productpage
                    pod-template-hash=57ffb6658c
                    version=v1
  Annotations:      prometheus.io/path: /metrics
                    prometheus.io/port: 9080
                    prometheus.io/scrape: true
  Status:           Running
  IP:               10.76.0.18
  IPs:
    IP:           10.76.0.18
  Controlled By:  ReplicaSet/productpage-v1-57ffb6658c
  Containers:
    productpage:
      Container ID:   containerd://31eb4fd882166381af91caf1e8fba8a6efd93d311697f3ba056893b792dcd4de
      Image:          docker.io/istio/examples-bookinfo-productpage-v1:1.20.2
      Image ID:       docker.io/istio/examples-bookinfo-productpage-v1@sha256:f3754ad97d9abfef9b92913dd57f3a6b5b1ede74a77b0fd699fa41984fa1d3a4
      Port:           9080/TCP
      Host Port:      0/TCP
      State:          Running
        Started:      Wed, 02 Oct 2024 12:05:13 +0200
      Ready:          True
      Restart Count:  0
      Environment:    <none>
      Mounts:
        /tmp from tmp (rw)
        /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-wtm8r (ro)
  Conditions:
```

When a user sends a request to the application, the following sequence of events occurs:

1. The request is received by the Service, which acts as a load balancer and forwards the request to one of the available pods running the application.
2. The pod processes the request by executing the application code within its container(s).
3. The pod generates a response, which is sent back to the Service.
4. The Service receives the response from the pod and forwards it back to the client (user).

This process is repeated for each incoming request, with the Service distributing the load across the available pods.

If the application needs to be scaled up or down to handle changes in traffic, the Deployment controller can create or remove pods accordingly. The Service automatically recognizes the new pods and includes them in the load balancing process, ensuring that user requests are distributed evenly across all available pods.

Additionally, if a pod fails or becomes unresponsive, the Deployment controller will automatically create a new pod to replace it, ensuring that the application remains highly available.

```
● oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ kubectl logs productpage-v1-57ffb6658c-x4gzd
  [2024-10-02 10:05:13 +0000] [1] [INFO] Starting gunicorn 22.0.0
  [2024-10-02 10:05:13 +0000] [1] [INFO] Listening at: http://[::]:9080 (1)
  [2024-10-02 10:05:13 +0000] [1] [INFO] Using worker: gevent
  [2024-10-02 10:05:13 +0000] [6] [INFO] Booting worker with pid: 6
  [2024-10-02 10:05:13 +0000] [7] [INFO] Booting worker with pid: 7
  [2024-10-02 10:05:13 +0000] [8] [INFO] Booting worker with pid: 8
  [2024-10-02 10:05:14 +0000] [9] [INFO] Booting worker with pid: 9
  [2024-10-02 10:05:14 +0000] [10] [INFO] Booting worker with pid: 10
  [2024-10-02 10:05:14 +0000] [11] [INFO] Booting worker with pid: 11
  [2024-10-02 10:05:14 +0000] [12] [INFO] Booting worker with pid: 12
  [2024-10-02 10:05:14 +0000] [13] [INFO] Booting worker with pid: 13
```

Changing the cluster's configuration by scaling up the ratings-v1 deployment to two Replica sets

```
oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ kubectl get deployment ratings-v1
NAME         READY   UP-TO-DATE   AVAILABLE   AGE
ratings-v1   1/1     1            1           42m
oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ kubectl scale deployment ratings-v1 --replicas=2
deployment.apps/ratings-v1 scaled
oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ kubectl get deployment ratings-v1
NAME         READY   UP-TO-DATE   AVAILABLE   AGE
ratings-v1   2/2     2            2           42m
```

Verification of the deployment has been updated:

```
oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ kubectl get pods
NAME                              READY   STATUS    RESTARTS   AGE
details-v1-6cd6d9df6b-dt9tj       1/1     Running   0          43m
productpage-v1-57ffb6658c-x4gzd   1/1     Running   0          43m
ratings-v1-794744f5fd-5tgz2       1/1     Running   0          76s
ratings-v1-794744f5fd-fhmqb       1/1     Running   0          43m
reviews-v1-67896867f4-j6clg       1/1     Running   0          43m
reviews-v2-86d5db4bd6-2rdsx       1/1     Running   0          43m
reviews-v3-77947c4c78-mhmh9       1/1     Running   0          43m
oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$
```

Issuing of kubectl logs <new pod ID> to verify it's up and running and listening on port 9080.

```
oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ kubectl logs ratings-v1-794744f5fd-5tgz2
Server listening on: http://0.0.0.0:9080
```

Scale ratings-v1 deployment back down to 1 replica:

```
oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ kubectl scale deployment ratings-v1 --replicas=1
deployment.apps/ratings-v1 scaled
oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ kubectl get pods
NAME                              READY   STATUS    RESTARTS   AGE
details-v1-6cd6d9df6b-dt9tj       1/1     Running   0          48m
productpage-v1-57ffb6658c-x4gzd   1/1     Running   0          48m
ratings-v1-794744f5fd-fhmqb       1/1     Running   0          48m
reviews-v1-67896867f4-j6clg       1/1     Running   0          48m
reviews-v2-86d5db4bd6-2rdsx       1/1     Running   0          48m
reviews-v3-77947c4c78-mhmh9       1/1     Running   0          48m
```

Lab Task 2 – Build and deploy a new image to GKE

I edited the bookinfo/src/productpage/templates/productpage.html file and inserted new text into the block title as shown:

```
</script>
{% block title %}My Modified BookInfo App{% endblock %}

<nav class="bg-gray-800">
  <div class="container mx-auto px-4 sm:px-6 lg:px-8">
    <div class="relative flex h-16 items-center justify-between">
      <a href="#" class="text-white px-3 py-2 text-lg font-medium" aria-current="page">BookInfo_Aiops</a>
      <div class="absolute inset-y-0 right-0 flex items-center pr-2 sm:static sm:inset-auto sm:ml-6 sm:pr-0">
        {% if user: %}
        <a href="#" class="group block flex-shrink-0">
          <div class="flex items-center">
```

Defined two environment variables:

```
oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ export BOOKINFO_HUB=oadewusi/productpage
oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ export BOOKINFO_TAG=latest
```

```
oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo/src$ ./build-services.sh
+++ dirname ./build-services.sh
++ cd .
++ pwd
+ SCRIPTDIR=/home/oadewusi/aiops/Assignments/LAB4/istio-master/samples/bookinfo/src
+ cd /home/oadewusi/aiops/Assignments/LAB4/istio-master/samples/bookinfo/src/../../..
+ h=oadewusi/productpage
+ t=latest
+ [[ oadewusi/productpage == \i\s\t\i\o ]]
+ [[ oadewusi/productpage == \d\o\c\k\e\r\.\i\o\/\i\s\t\i\o ]]
+ [[ '' == \t\r\u\e ]]
+ env TAGS=latest HUB=oadewusi/productpage docker buildx bake -f samples/bookinfo/src/docker-bake.hcl
[+] Building 28.7s (1/2)                                                   docker-container:priceless_nash
 => [internal] load local bake definitions                                                            0.0s
 => => reading samples/bookinfo/src/docker-bake.hcl 2.49kB / 2.49kB                                   0.0s
 => [internal] booting buildkit                                                                      28.7s
 => => pulling image moby/buildkit:buildx-stable-1                                                   28.7s
```

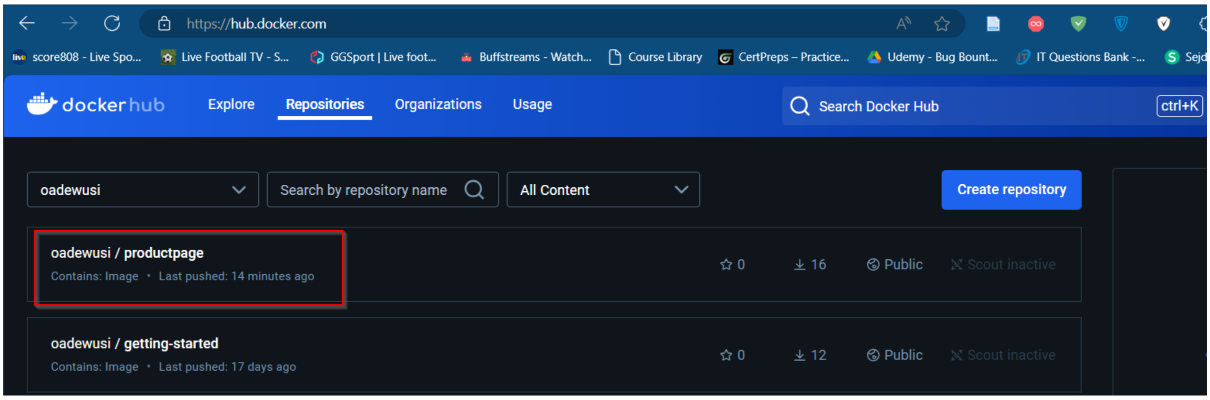
I used docker build -t $BOOKINFO_HUB:$BOOKINFO_TAG . to build the new image

```
oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo/src/productpage$ docker build -t $BOOKINFO_HUB:$BOOKINFO_TAG .
[+] Building 60.4s (18/18) FINISHED                                                    docker:default
```

Inspecting the image built with docker image ls and pushing it to docker hub:

```
oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo/src/productpage$ docker image ls
REPOSITORY             TAG      IMAGE ID       CREATED        SIZE
oadewusi/productpage   latest   d7f50732c9b3   2 hours ago    190MB
oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo/src/productpage$ docker login
Authenticating with existing credentials...
Login Succeeded
oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo/src/productpage$ docker push oadewusi/productpage:latest
```

Verification of the image successfully pushed to docker hub repo:



I edited the bookinfo.yaml to switch from the default image to my new image

```
! bookinfo.yaml ×
istio-master > samples > bookinfo > platform > kube > ! bookinfo.yaml
spec:
  template:
    metadata:
      labels:
        version: v1
    spec:
      serviceAccountName: bookinfo-productpage
      containers:
      - name: productpage
        image: docker.io/oadewusi/productpage:latest
        imagePullPolicy: Always
        ports:
        - containerPort: 9080
        volumeMounts:
        - name: tmp
          mountPath: /tmp
      volumes:
      - name: tmp
        emptyDir: {}
---
```

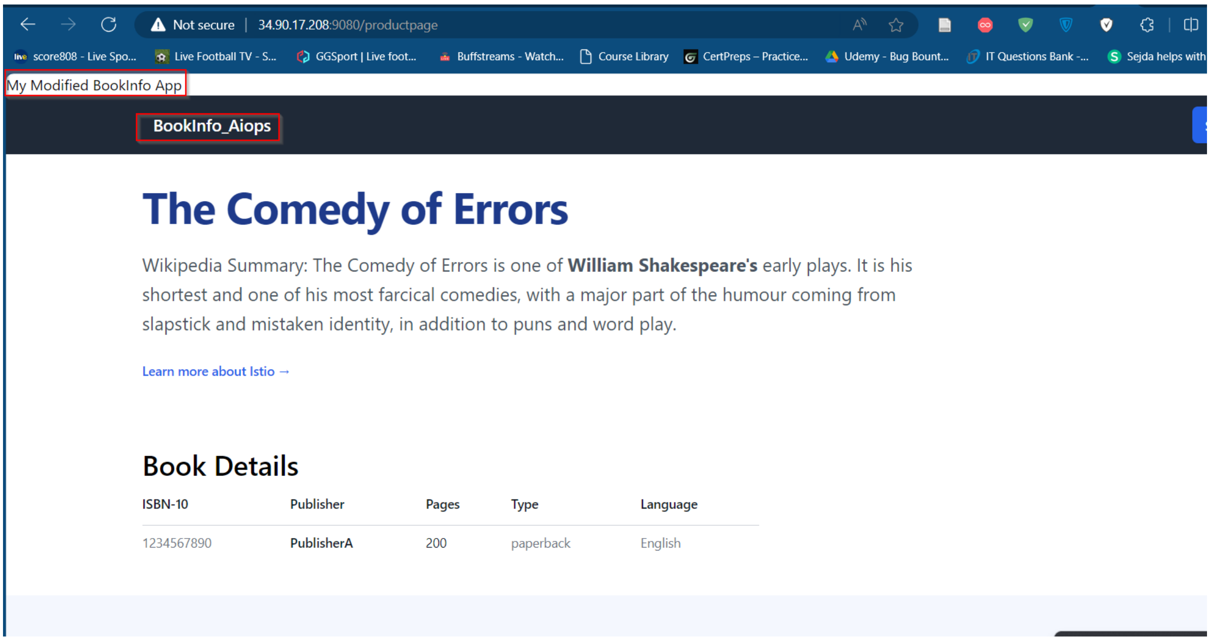
Redeploy my application by re-applying the Kubernetes.yaml file:

```
oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ kubectl apply -f platform/kube/bookinfo.yaml
service/details unchanged
serviceaccount/bookinfo-details unchanged
deployment.apps/details-v1 unchanged
service/ratings unchanged
serviceaccount/bookinfo-ratings unchanged
deployment.apps/ratings-v1 unchanged
service/reviews unchanged
serviceaccount/bookinfo-reviews unchanged
deployment.apps/reviews-v1 unchanged
deployment.apps/reviews-v2 unchanged
deployment.apps/reviews-v3 unchanged
service/productpage unchanged
serviceaccount/bookinfo-productpage unchanged
deployment.apps/productpage-v1 configured
```

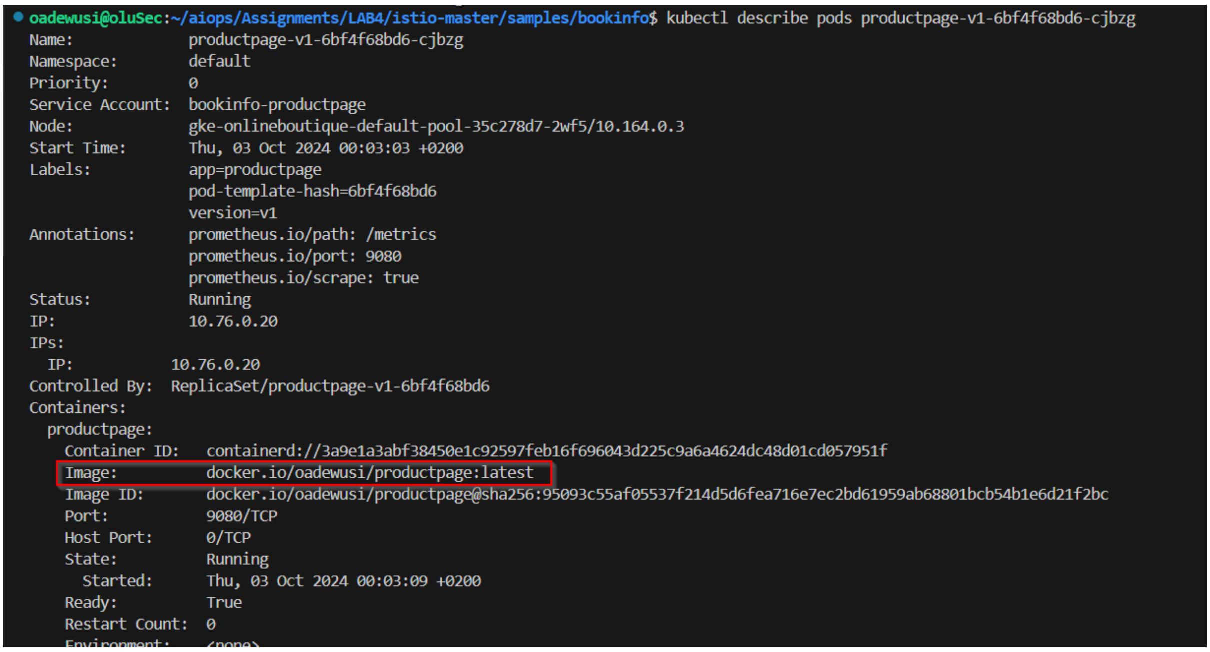
Then issued a kubectl get pod command to view the updated productpage:

```
oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ kubectl get pods
NAME                              READY   STATUS    RESTARTS   AGE
details-v1-6cd6d9df6b-dt9tj       1/1     Running   0          11h
productpage-v1-6bf4f68bd6-cjbzg   1/1     Running   0          62s
ratings-v1-794744f5fd-fhmqb       1/1     Running   0          11h
reviews-v1-67896867f4-j6clg       1/1     Running   0          11h
reviews-v2-86d5db4bd6-2rdsx       1/1     Running   0          11h
reviews-v3-77947c4c78-mhmh9       1/1     Running   0          11h
```

Reloaded the webpage to verify my new changes has been applied:



Then issued a describe pod command to check the new image attached:



Lab Task 3 – Monitor GKE with Prometheus/Grafana

Added Prometheus monitoring and a Grafana panel to track requests to the bookinfo ProductPage and edited the Prometheus config file in the sandbox "containers" directory for Prometheus and to add an entry for Bookinfo:

```
! config.yml ×    productpage.py
istio-master > samples > bookinfo > containers > prometheus > ! config.yml
scrape_configs:
  - job_name: "app_one"
    static_configs:
      - targets:
        - "app_one:8000"
  - job_name: "monitor_app"
    static_configs:
      - targets:
        - "monitor_app:8003"
  - job_name: "bookinfo"
    static_configs:
      - targets: ['34.90.17.208:9080']
```

Examined the Python source code to the BookInfo product page to verify the metric:

```
! config.yml    productpage.py ×
istio-master > samples > bookinfo > src > productpage > productpage.py

service_dict = {
    "productpage": productpage,
    "details": details,
    "reviews": reviews,
}

request_result_counter = Counter('request_result', 'Results of requests', ['destination_app', 'response_code'])
```

Bringing up my compose file:

```
oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$ docker compose up -d
 [+] Running 26/20
  ✓ postgres_exporter Pulled
  ✓ node_exporter Pulled
  ✓ push_gateway Pulled
  ✓ postgres Pulled
```

Verification of the containers created:

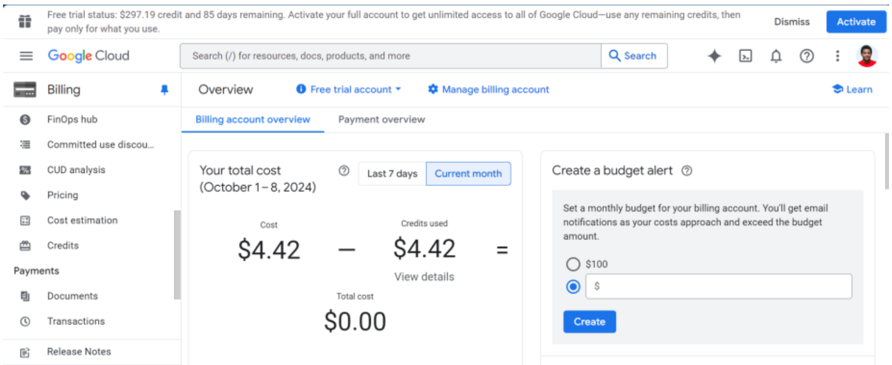
```
 ✓ Network bookinfo_default             Created
 ✓ Container prometheus_node_exporter   Started
 ✓ Container prometheus_push_gateway    Started
 ✓ Container grafana                    Started
 ✓ Container prometheus                 Started
 ✓ Container postgres                   Started
 ✓ Container postgres_exporter          Started
oadewusi@oluSec:~/aiops/Assignments/LAB4/istio-master/samples/bookinfo$
```

Navigated back to Grafana and to add the visualization for the product page metric. Hitting the product page a number of times to generate some manual load.

request_result_created metric and request_result_total metic for 5 mins

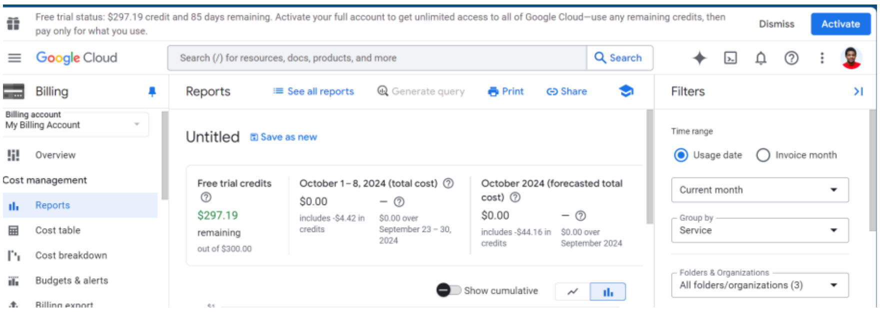
request_result_created metric and request_result_total metic for 10 mins

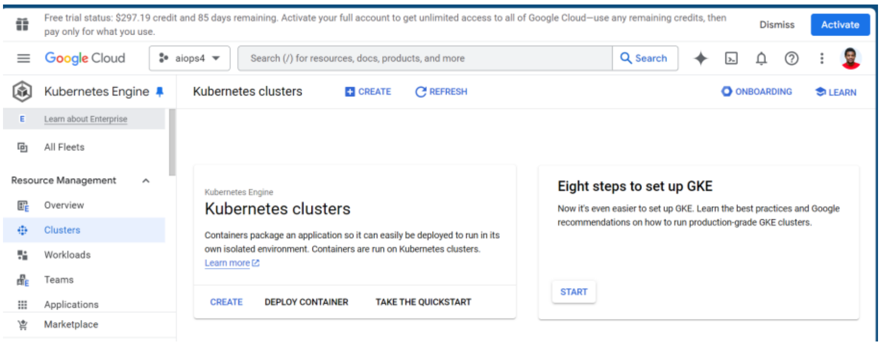
request_result_created metric and request_result_total metic for 15 mins



Service	Cost	Discounts	Promotions and others	Subtotal	% Change
Compute Engine	$2.29	–	–	$2.29	New
Kubernetes Engine	$1.42	–	–	$1.42	New
Networking	$0.55	–	–	$0.55	New
Cloud Monitoring	$0.15	–	–	$0.15	New

Subtotal	$4.42
Tax	–
Filtered total	$4.42

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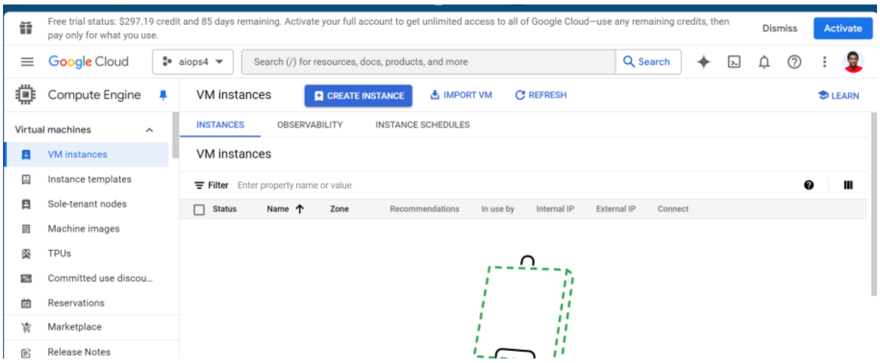
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External IP
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